1. Which metric is used to adjust spreads for embedded prepayment options in MBS?

   A) Nominal spread

   B) Option Adjusted Spread

   C) Forward Cashflow Spread

   D) Treasury-SOFR spread

2. Which concept captures the tendency of MBS prices to fall faster when rates decline due to faster prepayments?

   A) Gamma hedging

   B) Negative Convexity

   C) Roll-down carry

   D) Basis risk

3. Which prepayment pathway refers to borrowers repaying part of the principal ahead of schedule without fully refinancing?

   A) Strategic default

   B) Curtailment

   C) Cash-out refinance

   D) Streamline modification

4. In the TBA market, which characteristic typically defines a trade rather than specific loan-level attributes?

   A) Individual borrower FICO scores

   B) Property ZIP codes

C) Generic pool parameters such as coupon and settlement

D) Appraisal comps

5. Home price appreciation can influence prepayment dynamics primarily by:

   A) Reducing liquidity in the TBA roll market

   B) Eligibility for cash-out refinancing

   C) Weakening the negative convexity of MBS at all rate levels

   D) Eliminating the need for credit enhancement

6. Seasonality in mortgage prepayments is most likely to result in:

   A) Uniform speeds across all months

   B) Higher prepayment speeds during peak home-moving seasons

   C) Higher winter prepayments due to faster underwriting

   D) No observable pattern in scheduled principal

7. In a simulation of a simple Brownian motion with drift, given by $dX(t) = \mu\, dt + \sigma\, dW(t)$, how does the dispersion (standard deviation) of $X(t)$ evolve with time t?

   A) σ
   B) σt
   C) $\sigma \sqrt{t}$
   D) σt^2

8. Let $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ be independent normal random variables. Define $Z = aX + bY$ where a and b are nonzero constants. Which statement correctly gives the distribution of Z?

A) $Z \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
B) $Z \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$
C) $Z \sim N(a\mu_1 + b\mu_2, (a\sigma_1 + b\sigma_2)^2)$
D) Z is not normal unless a = b = 1

9. Under an inverted curve scenario, where the 2Y swap rate is higher than 5Y and the 5Y swap rate is higher than 10Y swap rate, consider the 1Y forward rate from 9Y to 10Y (the 9Y→10Y forward). Relative to the current 10Y par swap rate, this 1Y forward will most likely be:

   A) Lower than the 10Y swap rate
   B) Equal to the 10Y swap rate
   C) Higher than the 10Y swap rate
   D) Indeterminate without any assumptions

10. Consider a European receiver swaption on a 5Y into 10Y swap (option expires in 5 years on a 10-year underlying swap). Which change is most likely to increase the receiver swaption's value, all else equal?

   A) Increase in the forward 10Y swap rate relative to the strike
   B) Decrease in implied volatility of the forward 10Y swap rate
   C) Decrease in the forward 10Y swap rate relative to the strike
   D) Reduction in time to expiry from 5 years to 4 years